

Site Suitability Analysis for Solar Power Plant Using GIS and Multi-Criteria Decision Analysis(MCDA) for Kalahandi District, Odisha.

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Abstract:

The Indian state of Odisha has immense potential to tap into solar energy as a reliable source of renewable energy production to fuel its energy demands. In Kalahandi district, which is dominated by agriculture, decentralised solar energy projects can provide energy security and an income source to farmers who are increasingly facing income uncertainties due to shifting climatic patterns that affect their crop yields. This study uses Multi-Criteria Evaluation on satellite remote sensing data of Kalahandi district to determine suitable sites for developing solar power projects. The factors that were taken into consideration include Global Horizontal Irradiance (GHI), slope, aspect, altitude, distance to roads, and distance to built-up areas, wherein fuzzy analysis was used to determine suitability. Constraints (where solar projects absolutely cannot be developed) were taken to be water bodies, trees, flooded vegetation, built-up areas, and the Karlapat Wildlife Sanctuary area, including a 10-km buffer zone around it, in accordance with the Eco-Sensitive Zone mandates of the Government of India.

Using the Analytical Hierarchy Process to assign weights to the factors, the Weighted Sum method was used in ArcGIS Pro to generate a suitability map classified into 5 zones: least, less, sufficiently, moderately, and highly suitable. The result of the map indicated that approx 4,448 km² of the district is suitable, with 1,195 km² (15.09% of the total area) emerging as highly suitable. GHI was found to be generally high across all tehsils, from which it can be inferred that conventional crystalline silicon photovoltaic panels can be used for projects in this region.

1. Introduction:

One of such countries that has emphasized the goal of raising its use of renewables in its agenda for the forthcoming years is India as a nation. During COP26, it was set that by 2030, it should attain 500 GW of non-fossil fuel capacity (PIB Headquarters, n.d.). One of our major accomplishments, even before achieving our goal of having 49% of our installed power generation capacity from non-fossil fuel resources (International Energy Agency, 2025), has come true. For instance, there are massive solar power projects built by our countries like Bhadla (2.25 GW), located in Rajasthan; Pavagada (2 GW), located in Karnataka; and Kurnool (1000 MW), located in Andhra Pradesh, among others (Soleos, 2025).